

Week-3: DAY 14 – Hands On – Keerthivasan R

Level-1: Problem 1 – Product Analysis Using Nested Queries

Scenario:

You are working as a database developer for an automobile retail company. Management wants to identify products that are priced higher than the average price of products in their respective categories.

Requirements

1. Retrieve product details (product_name, model_year, list_price).
2. Compare each product's price with the average price of products in the same category using a nested query.
3. Display only those products whose price is greater than the category average.
4. Show calculated difference between product price and category average.
5. Concatenate product name and model year as a single column (e.g., 'ProductName (2017)').

Technical Constraints

- Use subquery in WHERE clause.
- Use aggregate function (AVG).
- Use string manipulation functions.
- Use arithmetic expressions for price difference calculation.

Expectations:

- Proper use of nested query.
- Correct calculation of average and difference.
- Clean and readable SQL query.

Learning Outcome

- Understand nested queries with aggregate functions.
- Perform calculations inside SELECT statement.
- Apply string concatenation in SQL.

QUERY:

```
USE EcommDb;
GO

SELECT
    p1.product_name + ' (' + CAST(p1.model_year AS VARCHAR(4)) + ')' AS Product_Info,
    p1.product_name,
    p1.model_year,
    p1.list_price,
    p1.list_price - (
        SELECT AVG(p2.list_price)
        FROM products p2
        WHERE p2.category_id = p1.category_id
    ) AS Price_Difference
FROM products p1
WHERE p1.list_price > (
    SELECT AVG(p2.list_price)
    FROM products p2
    WHERE p2.category_id = p1.category_id
);
```

OUTPUT:

	Product_Info	model_year	list_price	Price_Difference
1	Electra Girl's Hawaii 1 (20-inch) - 2015/2016 (2016)	2016	299.99	12.203390
2	Haro Downtown 16 - 2017 (2017)	2017	329.99	42.203390
3	Trek Precaliber 24 (21-Speed) - Girls - 2017 (2017)	2017	349.99	62.203390
4	Electra Girl's Hawaii 1 16" - 2017 (2017)	2017	299.99	12.203390
5	Electra Moto 3i (20-inch) - Boy's - 2017 (2017)	2017	349.99	62.203390
6	Electra Savannah 3i (20-inch) - Girl's - 2017 (2017)	2017	349.99	62.203390
7	Electra Straight 8 3i (20-inch) - Boy's - 2017 (2017)	2017	489.99	202.203390
8	Electra Sugar Skulls 1 (20-inch) - Girl's - 2017 (2017)	2017	299.99	12.203390
9	Electra Townie 8 EQ (20-inch) - Boys' - 2017 (2017)	2017	489.99	202.203390
10	Electra Townie 7D (20-inch) - Boys' - 2017 (2017)	2017	339.99	52.203390
11	Strider Strider 20 Sport - 2018 (2018)	2018	289.99	2.203390
12	Trek Superfly 20 - 2018 (2018)	2018	399.99	112.203390
13	Trek Precaliber 20 6-speed Boy's - 2018 (2018)	2018	289.99	2.203390
14	Trek Precaliber 20 6-speed Girl's - 2018 (2018)	2018	289.99	2.203390
15	Trek Precaliber 24 (7-Speed) - Boys - 2018 (2018)	2018	319.99	32.203390
16	Trek Precaliber 24 21-speed Boy's - 2018 (2018)	2018	369.99	82.203390
17	Trek Precaliber 24 21-speed Girl's - 2018 (2018)	2018	369.99	82.203390
18	Trek Precaliber 24 7-speed Girl's - 2018 (2018)	2018	319.99	32.203390
19	Trek Superfly 24 - 2017/2018 (2018)	2018	489.99	202.203390
20	Electra Cruiser 7D (24-Inch) Ladies' - 2016/2018 (2018)	2018	319.99	32.203390
21	Electra Heartchya 1 (20-inch) - Girl's - 2018 (2018)	2018	319.99	32.203390
22	Electra Savannah 1 (20-inch) - Girl's - 2018 (2018)	2018	319.99	32.203390
23	Electra Straight 8 1 (20-inch) - Boy's - 2018 (2018)	2018	389.99	102.203390
24	Electra Superbolt 1 20" - 2018 (2018)	2018	319.99	32.203390
25	Electra Superbolt 3i 20" - 2018 (2018)	2018	369.99	82.203390
26	Electra Sweet Ride 1 (20-inch) - Girl's - 2018 (2018)	2018	319.99	32.203390
27	Electra Sweet Ride 3i (20-inch) - Girls' - 2018 (2018)	2018	369.99	82.203390
28	Electra Tiger Shark 1 (20-inch) - Boys' - 2018 (2018)	2018	319.99	32.203390
29	Electra Tiger Shark 3i (20-inch) - Boys' - 2018 (2018)	2018	369.99	82.203390
30	Electra Treasure 1 20" - 2018 (2018)	2018	319.99	32.203390
31	Electra Treasure 3i 20" - 2018 (2018)	2018	369.99	82.203390
32	Electra Townie Balloon 3i EQ Ladies' - 2018 (2018)	2018	799.99	117.866667
33	Electra Townie Balloon 7i EQ - 2018 (2018)	2018	899.99	217.866667
34	Electra Townie Go! 8i - 2017/2018 (2018)	2018	2599.99	1917.866667
35	Electra Townie Balloon 3i EQ - 2017/2018 (2018)	2018	749.99	67.866667
36	Electra Townie Balloon 7i EQ Ladies' - 2017/2018 (2018)	2018	899.99	217.866667
37	Electra Townie Balloon 8D EQ - 2016/2017/2018 (2018)	2018	749.99	67.866667
38	Electra Townie Balloon 8D EQ Ladies' - 2016/2017/2018 (20...	2018	749.99	67.866667
39	Electra Townie Commute 27D - 2018 (2018)	2018	899.99	217.866667
40	Electra Townie Commute 27D Ladies - 2018 (2018)	2018	899.99	217.866667
41	Electra Townie Commute 8D - 2018 (2018)	2018	749.99	67.866667
42	Electra Townie Commute 8D Ladies' - 2018 (2018)	2018	699.99	17.866667
43	Sun Bicycles Brickell Tandem 7 - 2017 (2017)	2017	749.99	19.577693
44	Electra Glam Punk 3i Ladies' - 2017 (2017)	2017	799.99	69.577693

45	Sun Bicycles Brickell Tandem CB - 2017 (2017)	2017	761.99	31.577693
46	Electra Amsterdam Fashion 7i Ladies' - 2017 (2017)	2017	1099.99	369.577693
47	Electra Tiger Shark 3i - 2018 (2018)	2018	899.99	169.577693
48	Electra Queen of Hearts 3i - 2018 (2018)	2018	749.99	19.577693
49	Electra Super Moto 8i - 2018 (2018)	2018	899.99	169.577693
50	Electra Straight 8 3i - 2018 (2018)	2018	909.99	179.577693
51	Electra Delivery 3i - 2016/2017/2018 (2018)	2018	959.99	229.577693
52	Electra Daydreamer 3i Ladies' - 2018 (2018)	2018	899.99	169.577693
53	Electra Koa 3i Ladies' - 2018 (2018)	2018	899.99	169.577693
54	Electra Morningstar 3i Ladies' - 2018 (2018)	2018	749.99	19.577693
55	Electra Relic 3i - 2018 (2018)	2018	849.99	119.577693
56	Electra Townie Balloon 8D EQ - 2016/2017/2018 (2018)	2018	749.99	19.577693
57	Electra Townie Balloon 8D EQ Ladies' - 2016/2017/2018 (2018)	2018	749.99	19.577693
58	Electra Townie Commute 27D Ladies' - 2018 (2018)	2018	899.99	169.577693
59	Electra Townie Commute 8D - 2018 (2018)	2018	749.99	19.577693
60	Electra White Water 3i - 2018 (2018)	2018	749.99	19.577693
61	Electra Townie Go! 8i - 2017/2018 (2018)	2018	2599.99	1869.577693
62	Electra Townie Commute Go! - 2018 (2018)	2018	2999.99	2269.577693
63	Electra Townie Commute Go! Ladies' - 2018 (2018)	2018	2999.99	2269.577693
64	Electra Townie Go! 8i Ladies' - 2018 (2018)	2018	2599.99	1869.577693
65	Electra Townie Balloon 3i EQ - 2017/2018 (2018)	2018	749.99	19.577693
66	Electra Townie Balloon 7i EQ Ladies' - 2017/2018 (2018)	2018	899.99	169.577693
67	Electra Townie Commute 27D - 2018 (2018)	2018	899.99	169.577693
68	Electra Amsterdam Fashion 3i Ladies' - 2017/2018 (2018)	2018	899.99	169.577693
69	Electra Amsterdam Royal 8i - 2017/2018 (2018)	2018	1259.90	529.487693
70	Electra Amsterdam Royal 8i Ladies' - 2018 (2018)	2018	1199.99	469.577693
71	Electra Townie Balloon 3i EQ Ladies' - 2018 (2018)	2018	799.99	69.577693
72	Electra Townie Balloon 7i EQ - 2018 (2018)	2018	899.99	169.577693
73	Trek Boone 5 Disc - 2018 (2018)	2018	3299.99	757.197000
74	Trek Boone 7 Disc - 2018 (2018)	2018	3999.99	1457.197000
75	Trek Crockett 7 Disc - 2018 (2018)	2018	2999.99	457.197000
76	Trek Boone 7 - 2017 (2017)	2017	3499.99	957.197000
77	Trek Boone Race Shop Limited - 2017 (2017)	2017	3499.99	957.197000
78	Trek Powerfly 8 FS Plus - 2017 (2017)	2017	4999.99	1718.333334
79	Trek CrossRip+ - 2018 (2018)	2018	4499.99	1218.333334
80	Trek XM700+ Lowstep - 2018 (2018)	2018	3499.99	218.333334
81	Trek XM700+ - 2018 (2018)	2018	3499.99	218.333334
82	Trek Powerfly 5 - 2018 (2018)	2018	3499.99	218.333334
83	Trek Powerfly 5 FS - 2018 (2018)	2018	4499.99	1218.333334
84	Trek Powerfly 5 Women's - 2018 (2018)	2018	3499.99	218.333334
85	Trek Powerfly 7 FS - 2018 (2018)	2018	4999.99	1718.333334
86	Trek Super Commuter+ 7 - 2018 (2018)	2018	3599.99	318.333334
87	Trek Super Commuter+ 8S - 2018 (2018)	2018	4999.99	1718.333334
88	Trek Fuel EX 8 29 - 2016 (2016)	2016	2899.99	1250.232667
89	Trek Slash 8 27.5 - 2016 (2016)	2016	3999.99	2350.232667

90	Trek Remedy 29 Carbon Frameset - 2016 (2016)	2016	1799.99	150.232667
91	Surly Karate Monkey 27.5+ Frameset - 2017 (2017)	2017	2499.99	850.232667
92	Trek Fuel EX 9.8 29 - 2017 (2017)	2017	4999.99	3350.232667
93	Trek Fuel EX 5 27.5 Plus - 2017 (2017)	2017	2299.99	650.232667
94	Trek Fuel EX 9.8 27.5 Plus - 2017 (2017)	2017	5299.99	3650.232667
95	Trek Remedy 9.8 - 2017 (2017)	2017	5299.99	3650.232667
96	Trek Fuel EX 8 29 - 2018 (2018)	2018	3199.99	1550.232667
97	Trek Fuel EX 7 29 - 2018 (2018)	2018	2499.99	850.232667
98	Surly Krampus Frameset - 2018 (2018)	2018	2499.99	850.232667
99	Surly Troll Frameset - 2018 (2018)	2018	2499.99	850.232667
100	Surly ECR 27.5 - 2018 (2018)	2018	1899.00	249.242667
101	Heller Bloodhound Trail - 2018 (2018)	2018	2599.00	949.242667
102	Trek Procaliber 6 - 2018 (2018)	2018	1799.99	150.232667
103	Heller Shagamaw GX1 - 2018 (2018)	2018	2599.00	949.242667
104	Trek Fuel EX 5 Plus - 2018 (2018)	2018	2249.99	600.232667
105	Trek Remedy 7 27.5 - 2018 (2018)	2018	2999.99	1350.232667
106	Trek Remedy 9.8 27.5 - 2018 (2018)	2018	4999.99	3350.232667
107	Trek Fuel EX 8 29 XT - 2018 (2018)	2018	3199.99	1550.232667
108	Trek Domane SLR 6 - 2018 (2018)	2018	4999.99	1824.632667
109	Trek Domane SL 8 Disc - 2018 (2018)	2018	5499.99	2324.632667
110	Trek Domane SLR 8 Disc - 2018 (2018)	2018	7499.99	4324.632667
111	Trek Emonda SL 7 - 2018 (2018)	2018	4499.99	1324.632667
112	Trek Domane SL 7 Women's - 2018 (2018)	2018	4999.99	1824.632667
113	Trek Domane SLR 6 Disc Women's - 2018 (2018)	2018	5499.99	2324.632667
114	Trek Domane SLR 9 Disc - 2018 (2018)	2018	11999.99	8824.632667
115	Trek Domane SL Frameset - 2018 (2018)	2018	6499.99	3324.632667
116	Trek Domane SL Frameset Women's - 2018 (2018)	2018	6499.99	3324.632667
117	Trek Emonda SLR 6 - 2018 (2018)	2018	4499.99	1324.632667
118	Trek Emonda SLR 8 - 2018 (2018)	2018	6499.99	3324.632667
119	Trek Domane SL 6 - 2018 (2018)	2018	3199.99	24.632667
120	Trek Domane ALR Disc Frameset - 2018 (2018)	2018	3199.99	24.632667
121	Trek Domane ALR Frameset - 2018 (2018)	2018	3199.99	24.632667
122	Trek Domane SLR Disc Frameset - 2018 (2018)	2018	3199.99	24.632667
123	Trek Domane SLR Frameset - 2018 (2018)	2018	3199.99	24.632667
124	Trek Madone 9 Frameset - 2018 (2018)	2018	3199.99	24.632667
125	Trek Domane SLR 6 Disc - 2018 (2018)	2018	5499.99	2324.632667
126	Trek Domane SL 6 Disc - 2018 (2018)	2018	3499.99	324.632667
127	Trek Domane SL 6 - 2017 (2017)	2017	3499.99	324.632667
128	Trek Silque SLR 7 Women's - 2017 (2017)	2017	5999.99	2824.632667
129	Trek Silque SLR 8 Women's - 2017 (2017)	2017	6499.99	3324.632667
130	Trek Domane SL Disc Frameset - 2017 (2017)	2017	3199.99	24.632667
131	Trek Domane SLR 6 Disc - 2017 (2017)	2017	5499.99	2324.632667
132	Trek Madone 9.2 - 2017 (2017)	2017	4999.99	1824.632667
133	Trek Checkpoint SL 6 - 2019 (2019)	2019	3799.99	624.632667
134	Trek Checkpoint ALR Frameset - 2019 (2019)	2019	3199.99	24.632667